

- 4 (a) energy = average force $\times$ extension B1
 = $\frac{1}{2} \times F \times x$ B1
 (Hooke's law) extension proportional to (applied) force B1
 hence $F = kx$ B1
 so $E = \frac{1}{2}kx^2$ A0 [4]
- (b) (i) correct area shaded B1 [1]
- (ii) 1.0 cm² represents 1.0 mJ or correct units used in calculation C1
 $E_s = 6.4 \pm 0.2$ mJ A2 [3]
 (for answer $> \pm 0.2$ mJ but $\leq \pm 0.4$ mJ, then allow 2/3 marks)
- (iii) arrangement of atoms / molecules is changed B1 [1]
- 5 (a) (i) distance (of point on wave) from rest / equilibrium position B1 [1]